STAT 339
TUTORIAL SET 3

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

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1. Define $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : xy >= 0 \right\}$. Decide if W is a vector space and prove your claim.
(Hint: W is the union of the first and third quadrants in the $xy-$plane).

2. Let $\mathbb{S}$ be the set of all sequences of real numbers: $S = \{\mathbf{x} = (x_1, x_2, ...)|x_i \in \mathbb{R}\}$.

   a. Show that $\mathbb{S}$ is a vector space.

   b. Consider the subset T of $\mathbb{S}$ to be the set of real sequences that are "eventually zero" :
   $$T = \{\mathbf{x} \in \mathbb{S}|\exists n \in \mathbb{N} : x_i = 0, \forall i \geq n\}.$$
   Decide if T is a subspace of $\mathbb{S}$ and prove your claim.

3. Determine if the vector $\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$ is in the span of the set $\left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$.

4. Consider the vector space V of real-valued functions of a single variable. Determine if the following sets are subspaces of V:

   a. H_e, the set of even functions in V.

   b. H_o, the set of odd functions in V.

5. Let W be the set of all vectors of the form $\left\{ \begin{bmatrix} a+2b \\ 2c-a \\ b+c \\ 4c \end{bmatrix} : a, b, c \in \mathbb{R}. \right\}$.
Show that W is a subspace of $\mathbb{R}^4$.
Hint: Write W as the span of a set of vectors in $\mathbb{R}^4$.

6. Determine if the following statements are true or false :

   a. If U and W are subspaces of V, then $U \bigcup W$ is a subspace of V.

   b. If U and W are subspaces of V, then $U \bigcap W$ is a subspace of V.

   c. All lines in $\mathbb{R}^2$ are subspaces of $\mathbb{R}^2$.

   d. The set consisting of the zero vector is a vector space.

e. $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$.

7. Show that the positive quadrant $Q = \{(x, y)|x, y > 0\} \subset \mathbb{R}^2$ forms a vector space if we define addition by :

$$(x_1, y_1) + (x_2, y_2) = (x_1 \times x_2, y_1 \times y_2)$$

and scalar multiplication by

$$t(x, y) = (x^t, y^t),$$

where $x_i, y_i \in \mathbb{R}$ for $i \in 1, 2$ and $t \in \mathbb{R}$.

8. Let $\mathbb{S} = \{(x_1, x_2, x_3)|x_i \in \mathbb{R}, 1 \leq i \leq 3\}$. For $(x_1, x_2, x_3), (y_1, y_2, y_3) \in \mathbb{S}$ and $\mu \in \mathbb{R}$, define vector addition and scalar multiplication by:

   a. $(x_1, x_2, x_3)+(y_1, y_2, y_3) = (y_1, x_2+y_2, y_3)$ and $\mu(x_1, x_2, x_3) = (\mu x_1, \mu x_2, \mu x_3)$.
   b. $(x_1, x_2, x_3)+(y_1, y_2, y_3) = (x_1+y_1, x_2+y_2, x_3+y_3)$ and $\mu(x_1, x_2, x_3) = (x_1, 1, x_3)$.

   In each of the above cases, determine whether or not $\mathbb{S}$ is a vector space over $\mathbb{R}$ under the given operations.

9. Decide whether the following subsets of the vector space $M_{23}(\mathbb{R})$ are subspaces.

   a. $\{\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} |a = 2c + 1\}$.
   b. $\{\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} |a + c = 0, b + d + f = 0\}$.
   c. $\{\begin{bmatrix} 0 & 1 & a \\ b & c & 0 \end{bmatrix}\}$.

10. Determine which of the following subsets of $P_2(\mathbb{R})$ are subspaces.The set of all polynomials of the form $a_2t^2 + a_1t + a_o$ for which:

    a. $a_0 = 0$.
    b. $a_0 = 1$.
    c. $a_1 + a_2 = a_0$.